

Clinical Risk Assessment Report: Patient 9

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 13.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.59x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 1.7%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: 0.87 [Top 16%] (+0.9%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: -0.54 [Avg (27th %ile)] (+0.8%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 63.0 [Avg (47th %ile)] (-1.7%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-1.3%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

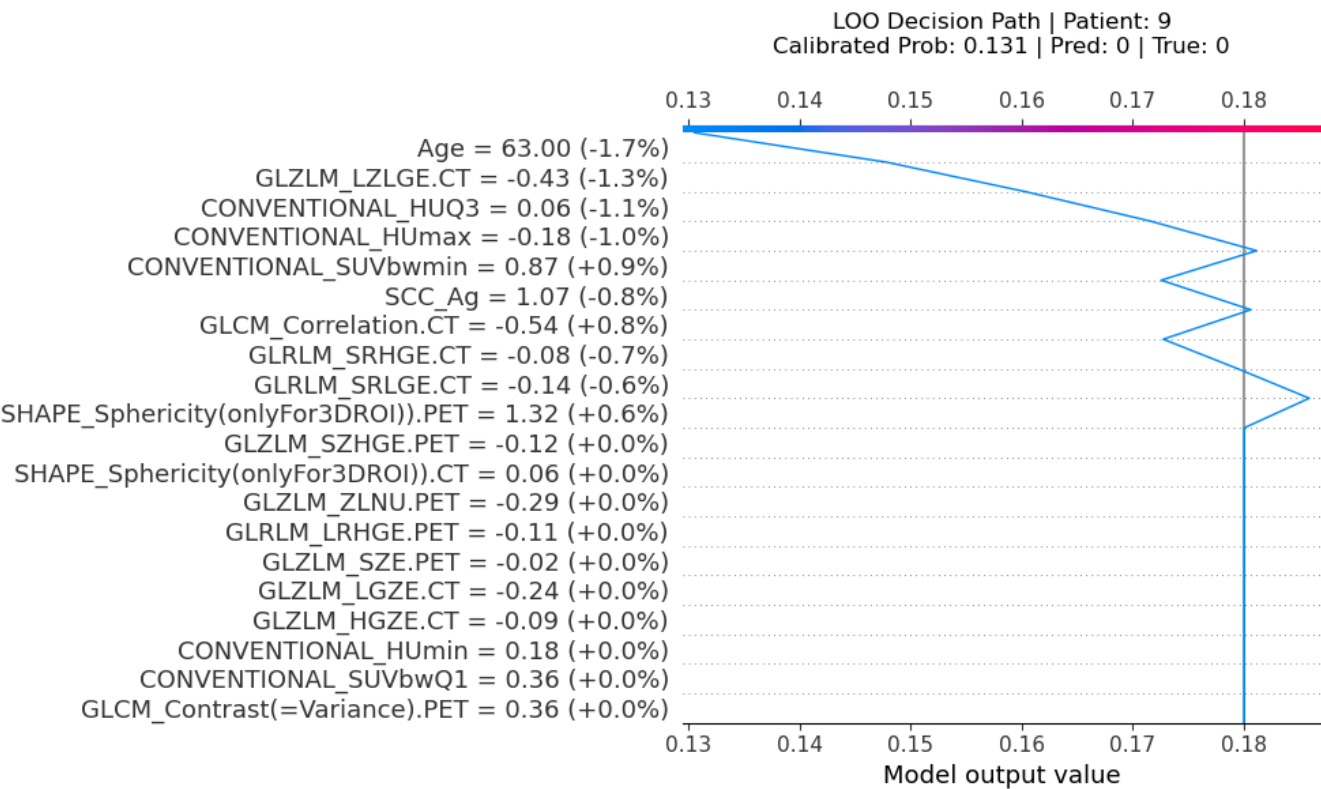
75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.06 [Avg (40th %ile)] (-1.1%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.18 [Avg (36th %ile)] (-1.0%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 1.07 [Top 13%] (-0.8%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: -0.08 [Avg (40th %ile)] (-0.7%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

